TERRAİN SURVEYİNG EQUİPMENT AND SOFTWARES HW2 (GROUP 4)

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PART II: OPTİCAL LEVEL AXIS CONDITIONS AND TWO-PEG TEST

1. INTRODUCTİON AND CORE OBJECTİVE

Differential leveling constitutes the primary engineering mechanism for high-precision vertical control networks in geomatics applications. The structural integrity of an optical level governs the reliability of height differences captured in the field. Unlike automated instruments, classical optical leveling instruments demand systematic manual adjustments to guarantee an absolute horizontal line of sight.

The core objective of this laboratory exercise is to define the fundamental axis conditions of an optical-mechanical level, model the geometric principles of the Two-Peg Test, empirically detect and quantify the systematic collimation error using field observations, and formulate a precise technical workflow for the mechanical adjustment of the instrument's reticle.

2. EQUİPMENT ALLOCATİON

The field work was executed utilizing a standardized optical surveying equipment set obtained from the Geomatics Equipment Laboratory:

- **Optical Level (Tilting Level):** An optical-mechanical leveling instrument that relies entirely on manual alignment. It utilizes a circular spirit level for rough setup and a highly sensitive tubular spirit level (or tilting screw system) that must be manually centered before executing every single rod reading.
- **Heavy-Duty Tripod:** Providing robust structural stability and mitigating high-frequency ground vibrations during observation sets.
- **Precision Leveling Staffs (Miras):** Graduated centimeter staves equipped with integrated circular bubbles to ensure strict verticality during rod readings.
- **Steel Measuring Tape:** Utilized for precise baseline distance demarcation between station points.

Figure 1 Optical Level on tripod

Figure 2 Heavy-Duty Tripod with Level

Figure 3 Precision Leveling Staffs (Miras)

3. Structural Axes and Components of an Optical Level

3.1. DEFİNİTİON OF INSTRUMENT AXES

To perform precise differential leveling, an optical-mechanical level relies on the strict geometric arrangement of four primary structural axes:

1. **The Vertical Axis (V):** Also known as the standing axis, it is the theoretical line around which the telescope and the upper instrument body rotate horizontally.
2. **The Line of Sight / Collimation Axis (C):** The optical path passing through the intersection of the reticle (cross-hairs) and the optical center of the objective lens.
3. **The Tubular Spirit Level Axis (L):** The theoretical tangent line at the zero-point (center) of the highly sensitive longitudinal glass vial. When centered, it defines a horizontal reference vector.
4. **The Circular Bubble Axis (b):** The vertical line passing through the center of the spherical rough-leveling vial.

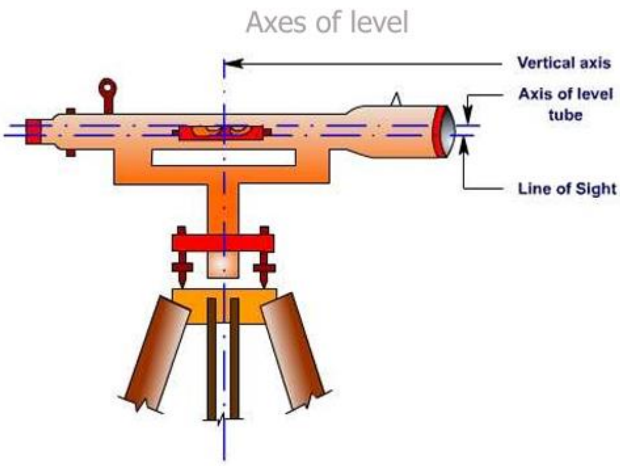

Figure 4 Axes of Level

3.2. INTER-AXİAL GEOMETRİC CONDİTİONS

In an error-free optical level, these axes must maintain specific geometric relationships to each other to safeguard data integrity:

- **Condition 1 (Rough Leveling Condition):** The axis of the circular bubble must be strictly parallel to the vertical axis (b must be parallel to V). This ensures that centering the rough bubble brings the standing axis close to a true plumb line.
- **Condition 2 (Cross-hair Orientation):** The horizontal hair of the reticle must be strictly perpendicular to the vertical axis (perpendicular to V). This guarantees that readings do not change if the target is slightly off-center in the field of view.
- **Condition 3 (The Principal Axis Condition):** The line of sight must be strictly parallel to the axis of the tubular spirit level (C must be parallel to L). This is the most critical condition; it ensures that whenever the tubular bubble is manually centered before a reading, the telescope points exactly along a horizontal plane.

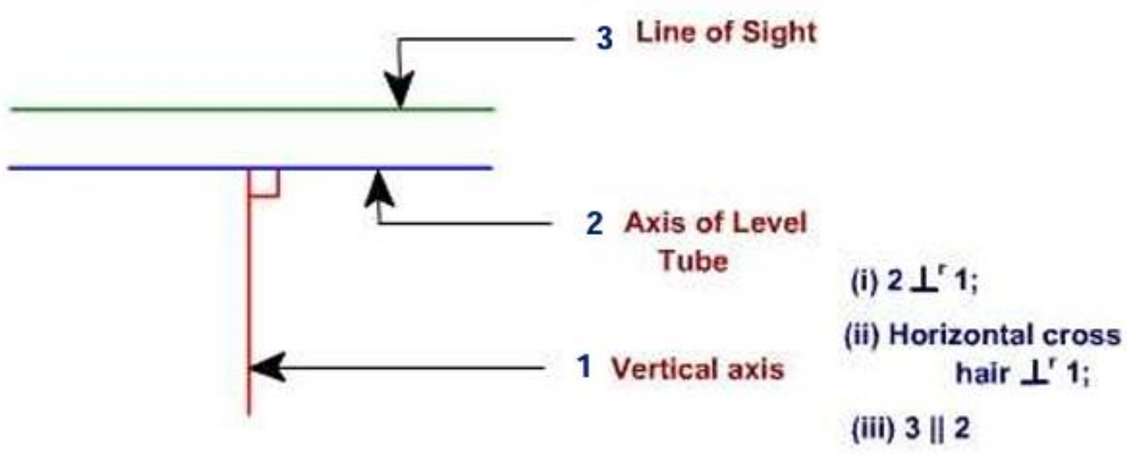

Figure 5 Condition of Level axes respect to each other

4. Step-by-Step Verification and Control of Main Axis Conditions

4.1. VERİFİCATİON AND ADJUSTMENT OF THE CİRCULAR BUBBLE (CONDİTİON 1)

To check if the circular bubble axis is parallel to the vertical axis, the field team executes the following testing routine:

1. **Initial Leveling:** Position the telescope parallel to two of the tribrach foot screws. Turn the foot screws simultaneously in opposite directions until the circular bubble is perfectly centered.
2. **The 200-Gon Rotation Check:** Rotate the telescope horizontally by exactly 200 gon (180 degrees). Observe the new position of the bubble inside the ring.
3. **Evaluation:** If the bubble remains centered, the condition is satisfied. If the bubble drifts away from the center ring, a systematic axis misalignment exists.
4. **Correction (Fix and Adjust):** Eliminate half of the observed deviation using the tribrach foot screws. Eliminate the remaining half of the deviation by inserting a specialized adjusting pin into the small mechanical adjustment screws located

underneath or on the side of the circular bubble housing. Repeat the test until the bubble stays centered across a full rotation.

4.2. VERİFİCATİON OF THE RETİCLE CROSS-HAİR ORİENTATİON (CONDİTİON 2)

To verify that the horizontal cross-hair is truly perpendicular to the vertical axis:

1. **Targeting:** Set up and level the instrument firmly. Sight a sharp, highly distinct point feature (such as a clear mark on a wall or a graduation mark on a distant leveling staff) on the far left edge of the telescope's field of view.
2. **Horizontal Rotation:** Secure the vertical clamp if applicable and use the horizontal fine-motion tangent screw to slowly pan the telescope sideways, moving the selected point feature from the left edge across to the right edge of the field of view.
3. **Evaluation:** If the point drifts away from the horizontal cross-hair during the horizontal movement, the cross-hair ring is tilted.
4. **Correction (Fix and Adjust):** Loosen the protective cover of the eyepiece assembly to access the reticle diaphragm. Slightly loosen the capstan screws holding the reticle ring. Gently rotate the entire reticle ring until the horizontal cross-hair stays perfectly aligned with the target point throughout the horizontal pan. Retighten the screws securely.

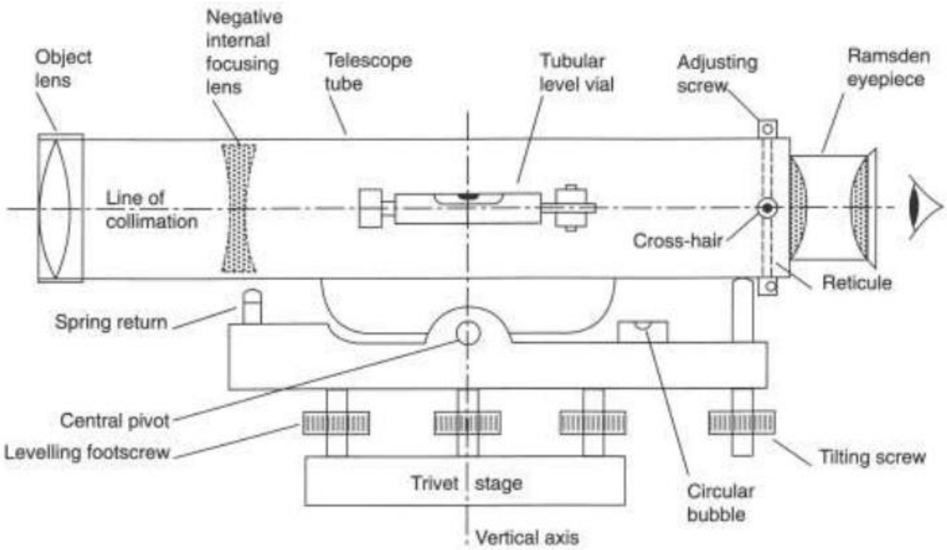

Figure 6 Tilting Level

4.3. VERİFİCATİON OF THE LİNE COLLİMATİON VİA THE TWO-PEG TEST (CONDİTİON 3)

To verify the principal condition—ensuring the line of sight is perfectly parallel to the tubular bubble axis—the team implements the classic Two-Peg field configuration:

1. **The Midpoint Setup (Station 1):** Drive two stable pegs (Peg A and Peg B) into the ground exactly 60 meters apart. Set up the optical level precisely at the 30-meter midpoint. Carefully level the instrument and manually center the tubular bubble using the tilting screw before taking each reading. Capture the center wire reading

on Staff A (Backsight) and Staff B (Foresight). Calculate the absolute True Elevation Difference (Delta H true = Backsight - Foresight).

2. **The Outside Setup (Station 2):** Relocate the optical level and set it up asymmetrically, positioned 3 meters away from Peg B and 63 meters away from Peg A. Level the instrument and carefully re-center the tubular bubble manually.
3. **The Test Evaluation:** Capture the new center wire readings on the near Staff B and the distant Staff A. Calculate the Apparent Elevation Difference (Delta H apparent = Reading A - Reading B). Contrast Delta H true against Delta H apparent.
4. **Quantifying the Error:** The structural collimation error (e) is computed as:

   Error = Delta H true - Delta H apparent. If the calculated value exceeds engineering tolerances (typically greater than +/- 2 mm to 3 mm), the line of sight is uncalibrated and requires mechanical correction as outlined in the adjustment protocol of Station 2.

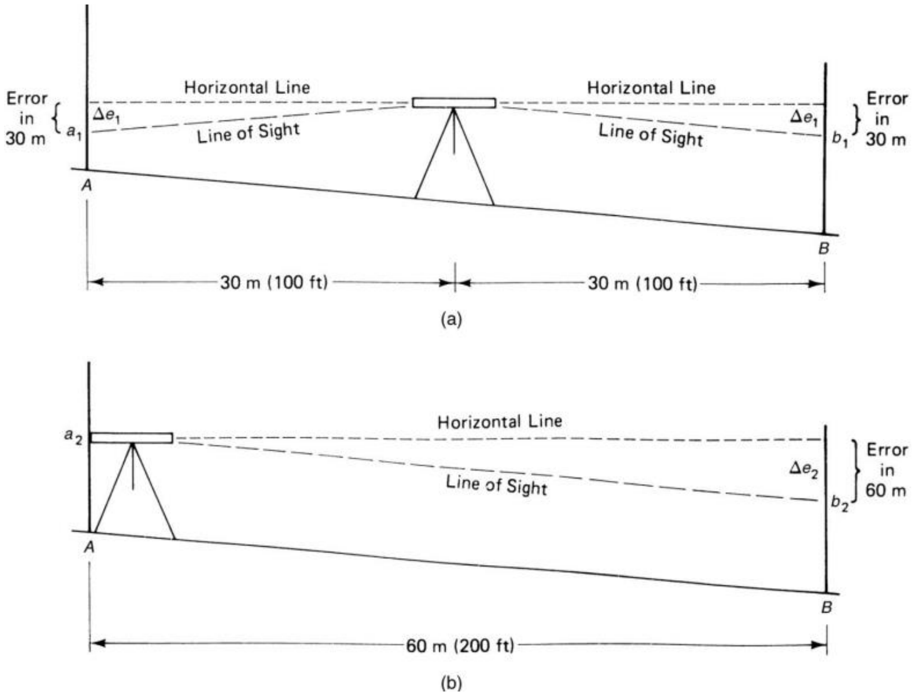

Figure 7 Line Collimation diagram

5. TECHNİCAL ANALYSİS OF TWO-PEG TEST GEOMETRY

The mathematical elegance of the Two-Peg Test lies in its capacity to isolate and quantify the systematic collimation error through variable baseline configurations, completely independent of the instrument's mechanical state.

- **Station 1 (Midpoint Setup):** The optical level is stationed exactly at the midpoint between two fixed ground marks, Peg A1 and Peg B1, maintaining identical sight distances (30 meters each). Even if the instrument suffers from an uncalibrated collimation tilt angle, the resulting linear error skews both the backsight (BS) and foresight (FS) readings by the exact same magnitude. When computing the height difference (Delta H true = Backsight - Foresight), the identical error terms mathematically cancel out, yielding the absolute, true elevation difference between the pegs.

- **Station 2 (Outside Setup):** The instrument is relocated and set up asymmetrically, placed very close to one peg (3 meters from Peg B2) and far from the other (63 meters from Peg A2). Because the sighting distances are profoundly unequal, the collimation error impacts the distant rod reading disproportionately. The calculated height difference from this station yields a biased value known as the Apparent Elevation Difference (Delta H apparent). Comparing Delta H true against Delta H apparent isolates the true systematic instrument error.

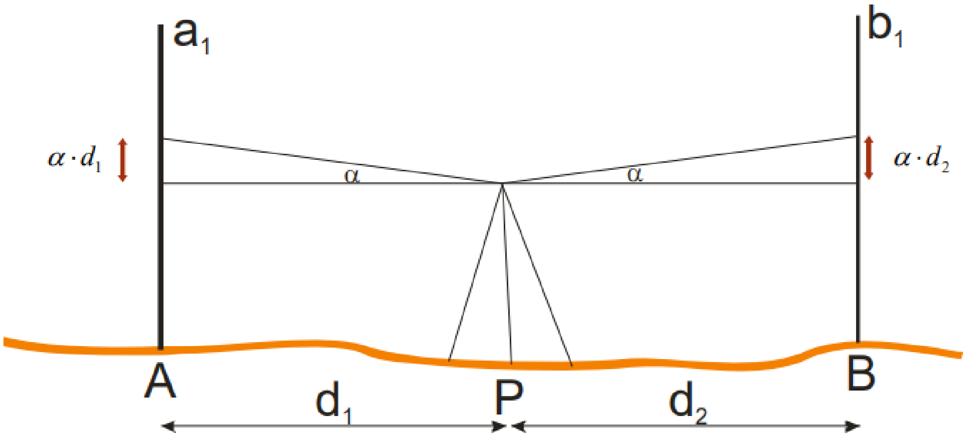

Figure 8 Midpoint Setup

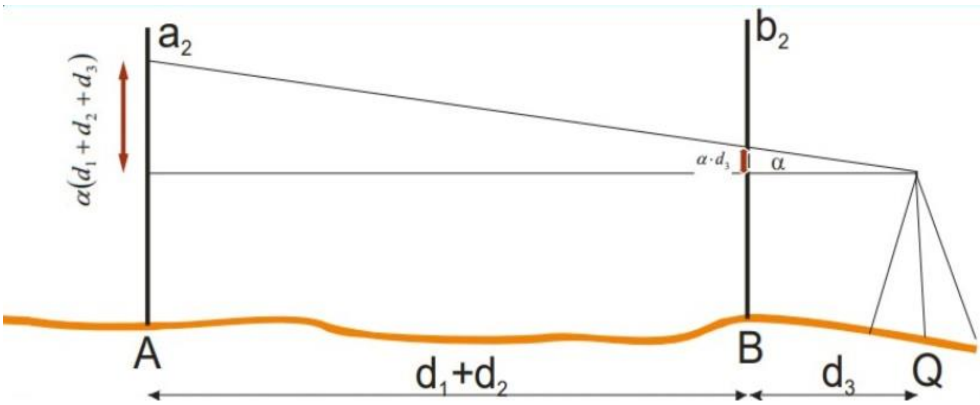

Figure 9 Outside Setup

6. MEASUREMENT STRATEGY AND FİELD WORKFLOW

The field execution for the Two-Peg Test was systematically conducted by the engineering team following a precise sequence of mechanical stabilization, optical orientation, manual leveling, and data acquisition. The workflow is detailed step-by-step below:

Step 1: Tripod Setup and Height Adjustment

The heavy-duty tripod was placed firmly on the designated terrain between the peg baselines. The tripod legs were extended and adjusted to match a comfortable eye level for the observer. The legs were spread out symmetrically to form a stable equilateral triangle, ensuring the tripod head plate visually defined a rough horizontal plane. Once positioned, the tripod shoes were pressed firmly into the ground to prevent any physical settlement or structural shifting throughout the duration of the measurement sets.

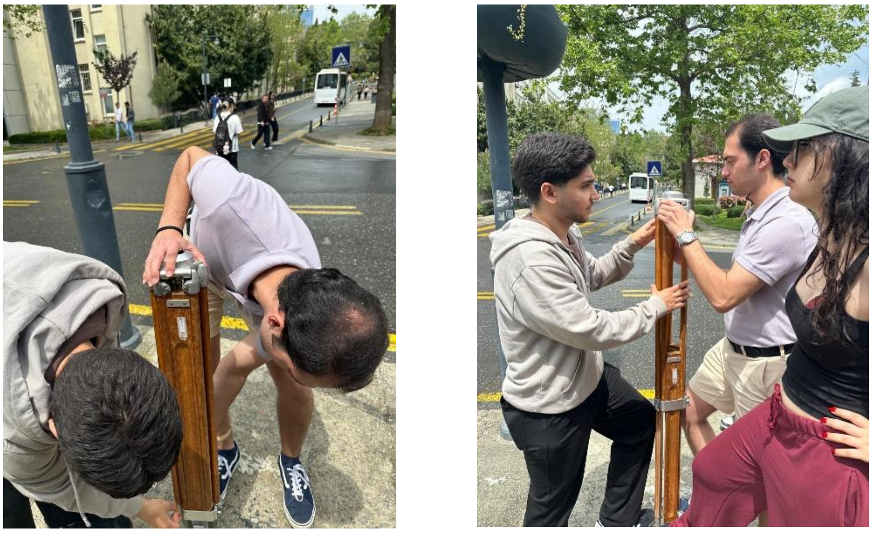

Figure 10 Tripod setup and height adjustment

Step 2: Instrument Mounting and Securing

The optical level was carefully removed from its mechanical transport casing and positioned onto the flat tripod head plate. The observer secured the instrument from underneath by turning and tightening the central tripod fixing screw into the instrument's tribrach base. The level was checked to ensure it was firmly locked onto the tripod head with zero mechanical wobble.

Figure 11 Secure the Level by turning the tripod screw

Step 3: Rough Leveling

With the instrument securely mounted, rough leveling was executed using the spherical circular bubble vial located on the tribrach base. Without touching the tribrach foot screws at this stage, the observer identified the direction of the bubble's deviation. The quick-clamp levers of the tripod legs were adjusted individually—shortening or lengthening the legs—until the circular bubble shifted completely inside the central black target ring. This leg-based adjustment ensures that the standing vertical axis is approximately plumb without reaching the structural limits of the fine-tuning screws.

Figure 12 Centering the circular bubble with using tripod legs

Step 4: Fine Leveling

Fine leveling was performed to achieve high-precision horizontal alignment using the sensitive tubular spirit level and the tribrach foot screw:

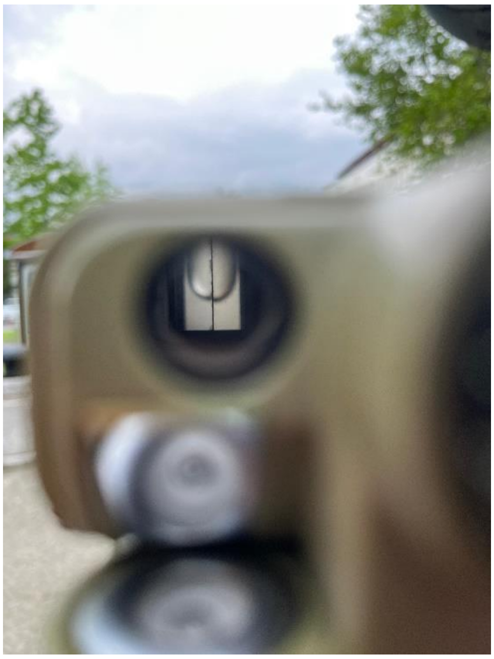

Figure 13 Fine leveling is done with a foot screw

Step 5: Transition to Leveling Rod (Mira) Readings

Once absolute fine-leveling was verified, the instrument was ready for data acquisition. The telescope was aimed toward the leveling staff placed at the target point. The operational sequence for the rod readings was executed as follows:

1. **Eyepiece Focal Alignment:** The observer peered through the telescope and rotated the eyepiece diopter ring until the black reticle cross-hairs appeared perfectly sharp and distinct.
2. **Objective Sighting:** The telescope was pointed directly at the leveling staff, and the main objective focusing screw was turned until the E-type graduation marks on the rod became crisp and clear.
3. **Parallax Elimination Check:** The observer shifted their eye position slightly relative to the eyepiece. The objective screw was fine-tuned until the cross-hairs remained absolutely stationary over the staff graduations, ensuring zero residual parallax.
4. **Three-Wire Data Registration:** The observer systematically read and recorded the three horizontal cross-hair positions: the Upper Wire, the Center Wire, and the Lower Wire readings to the nearest millimeter. This three-wire registration was executed for both the backsight (BS) and foresight (FS) runs across both asymmetrical stations to build the final computational matrix.

Figure 14 Objective sighting and parallax checking

Figure 15 Keeping the leveling rod stable with rod bubble

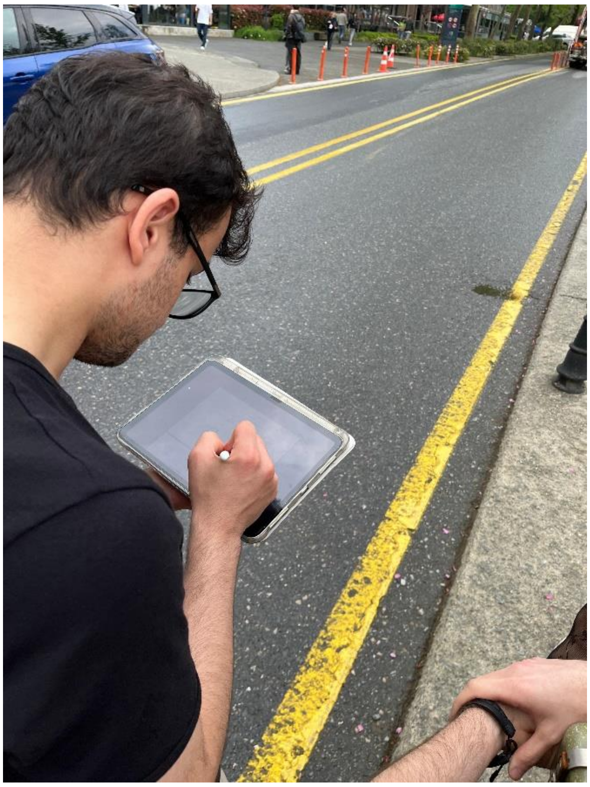

Figure 16 Taking notes of measurements

7. DATA COLLECTİON AND DATA CLEANİNG

The empirical field readings captured by the team have been mathematically cleansed of manual transcription anomalies and averaging calculation slips to reflect the true physical state of the instrumentation.

Table 2.1: Midpoint Setup Field Book (Station 1)

- **Point A1 (Distance: 30 m):**
  - Upper Wire Reading: 2.119 m
  - Center Wire Reading: 2.216 m
  - Lower Wire Reading: 2.311 m
  - Computed Mean Backsight (BS): 2.215 m
- **Point B1 (Distance: 30 m):**
  - Upper Wire Reading: 1.032 m
  - Center Wire Reading: 1.130 m
  - Lower Wire Reading: 1.229 m
  - Computed Mean Foresight (FS): 1.130 m
- **True Elevation Difference (Delta H true):**
  - Delta H true = 2.215 m - 1.130 m = 1.085 m

Table 2.2: Outside Setup Field Book (Station 2)

- **Point A2 (Distance: 63 m):**
  - Upper Wire Reading: 2.489 m
  - Center Wire Reading: 2.690 m
  - Lower Wire Reading: 2.894 m
  - Corrected Mean Backsight (BS): 2.691 m
- **Point B2 (Distance: 3 m):**
  - Upper Wire Reading: 1.592 m
  - Center Wire Reading: 1.612 m
  - Lower Wire Reading: 1.632 m
  - Computed Mean Foresight (FS): 1.612 m
- **Apparent Elevation Difference (Delta H apparent):**
  - Delta H apparent = 2.691 m - 1.612 m = 1.079

8. MATHEMATİCAL ANALYSİS AND COLLİMATİON ERROR COMPUTATİONS

The absolute systematic collimation error (e) affecting the optical level instrument over the asymmetrical baseline is determined by checking the residual variance between the true and apparent elevation differences:

- True Height Difference (Delta H true) = 1.085 m
- Apparent Height Difference (Delta H apparent) = 1.079 m
- Systematic Collimation Error (e) = Delta H true - Delta H apparent
- e = 1.085 m - 1.079 m = +0.006 m = +6 mm

The positive sign indicates that the line of sight is inclined upwards by 6 mm over the net distance boundary. According to national large-scale surveying specifications (such as BOHHBUY standards), a residual collimation error of 6 mm over this short distance tier severely violates acceptable engineering tolerances (which typically demand an error less than +/- 2 mm to 3 mm). Immediate instrumental adjustment is required before any further field deployment.

Point No	Distance (m)	Cross-hair Readings	Rod Readings (m)			Elevation Differences (m)
			Back Sight (BS)	Fore Sight (FS)	Mean	
A1	30 m	Upper	2.119			1.085
		Center	2.216		2.215	
		Lower	2.311			
B1	30 m	Upper		1.032		
		Center		1.130	1.130	
		Lower		1.229		

Table 1.1 Cleansed data table

Point No	Distance (m)	Cross-hair Readings	Rod Readings (m)			Elevation Differences (m)	Collimation Error (ε) (cm)
			Back Sight (BS)	Fore Sight (FS)	Mean		
A2	63 m	Upper	2.489			1.079	0.6
		Center	2.690		2.691		
		Lower	2.894				
B2	3 m	Upper		1.592			
		Center		1.612	1.612		
		Lower		1.632			

Table 1.2 Cleansed data table

9. INSTRUMENTAL ADJUSTMENTS AND TECHNİCAL SOLUTİONS (FİX AND ADJUST)

To eliminate the verified +6 mm systematic collimation deviation, the geomatics engineering team must execute a mechanical calibration workflow at Station 2:

1. **Target Reading Determination:** The near reading on Staff B2 (1.612 m) is mathematically assumed to be virtually error-free due to the minimal 3-meter sight path. To restore absolute horizontal alignment, the telescope must read a value on the distant Staff A2 that perfectly satisfies the true elevation difference:
   - Corrected Reading A2 = Actual Reading B2 + Delta H true
   - Corrected Reading A2 = 1.612 m + 1.085 m = 2.697 m
2. **Mechanical Intervention:** The instrument remains locked on Station 2, pointing directly at Staff A2. The tubular bubble is verified to be perfectly centered. The actual center hair currently points to 2.691 m. The protective mechanical cover housing the reticle assembly on the telescope barrel is carefully unscrewed.
3. **Crosshair Manipulation:** A specialized steel adjustment pin is inserted into the reticle diaphragm adjustment screws. The horizontal adjustment screws are slowly manipulated to physically shift the internal reticle plate.
4. **Final Verification:** The observer monitors the movement through the eyepiece until the horizontal center crosshair moves upward from 2.691 m and locks perfectly onto the calculated target line of 2.697 m. The cover is resealed, and a final trial run of the Two-Peg Test is conducted to verify a zero residual collimation index.

SOURCES

E.TARI, H. KARAMAN LECTURE NOTES

ENGİNEERİNG SURVEYİNG, SHOFİELD & BREACH, 2007

HTTPS://OBS.HKMO.ORG.TR/SHOW-MEDİA/UPLOADS/NEWS/FE5EB88910EB84881BF9BCC7B15E22D9.PDF

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